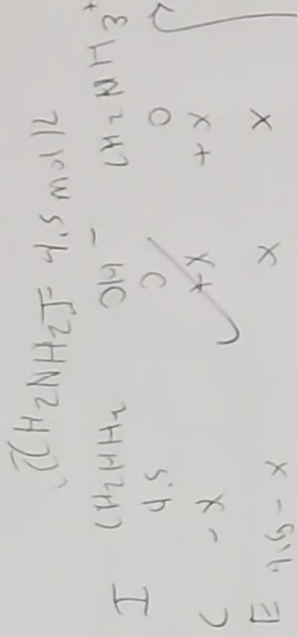
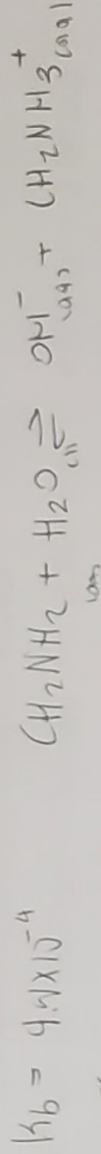


If the K_b for methylamine, $\text{CH}_3\text{NH}_2(\text{aq})$, is 4.4×10^{-4} , calculate the pH of a 4.5 mol/L solution of methylamine. (5 marks)



what's the number?

$$\sqrt{K_b} \cdot 4.5 = x$$

$$0.044477 = x$$

$$\text{pOH} = -\log(0.044477)$$

$$\text{pOH} = 1.3$$

$$\text{pH} = 14 - 1.3 = 12.7$$

4.5

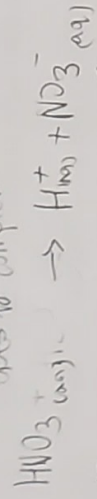
2 decimal places

$$4.4 \times 10^{-4} K_b = \frac{x^2}{(4.5-x)}$$

4. What is the pH of a solution when 25.00 mL of 0.20 mol/L $\text{HNO}_3(\text{aq})$ is titrated with 40.00 mL of 0.20 mol/L $\text{Ba}(\text{OH})_2$? (6 marks)

Strong base

goes to completion



goes to completion

HNO_3

limiting reagent, less

mmol/L

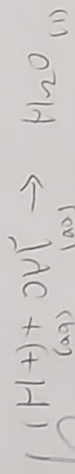
$$n_{\text{HNO}_3} = 5 \text{ mmol}$$

$$n_{\text{Ba}(\text{OH})_2} = 1 \text{ mmol}$$

using molar ratio

11 mmol/L

OH



$$16 - 5 = 11 \text{ mmol/L OH}$$

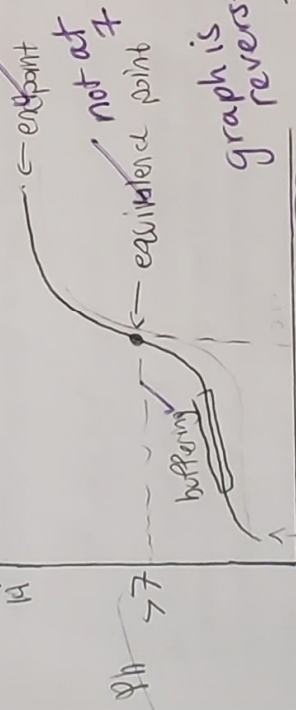
$$\text{pOH} = \frac{11 \text{ mmol}}{65 \text{ mL}} = 0.169$$

$$\text{pH} = 13.22$$

very basic solution

COMMUNICATION (10 MARKS)

1. You are given a solution of $\text{NaOH}(\text{aq})$ that has a concentration of 0.10 M. You titrate the solution with 0.10 M $\text{HC}_2\text{H}_3\text{O}_2(\text{aq})$. Sketch a detailed titration curve for the titration in the axis provided below, showing the equivalence point and its pH relative to $\text{pH}=7$. Label the names of the 3 distinct regions. (4 marks)



1. The solution first starts with a pH slightly higher than a strong acid because it is a weak acid.

2. Buffering occurs slowly and pH rises through not as flat as a strong acid + strong base.

3. Equivalence point is reached and is more basic as we added strong base only neutralize that acid.

4. Equivalence is reached at a high pH.

2

2. Using Le Chatelier's principle for the endothermic reaction below, which direction will the equilibrium shift when exposed to the changes below? (3 marks)

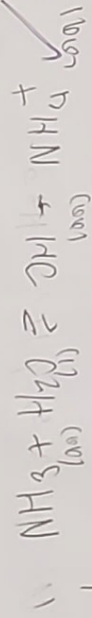


Stress	Direction of Shift
Addition of vanadium (III) ion	Right ✓
Br_2 is removed	Left ✓
addition of Ne to the container	No change ✓
Br^- is added	Left ✓
The system is cooled	Left ✓
Pressure is decreased	No change ✓

3. A buffer solution contains aqueous ammonia and ammonium chloride.

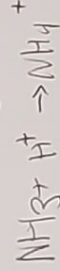
- a) Write an equation for the equilibrium system. (1 mark)
 b) In which direction does the equilibrium shift if hydrochloric acid is added? Why? (2 marks)

a) NH_3 and NH_4Cl dissolve and we have



b) If HCl is added more H^+ is introduced, that means the base

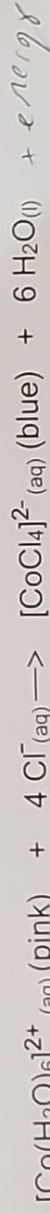
NH_3 will react w/ H^+ this shifting to the right



$\text{H}^+ + \text{OH}^-$ makes H_2O Shift R.

APPLICATION (10 MARKS)

1. Cobalt(II) chloride can form two complexes when dissolved in water: $[\text{Co}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ which is pink in colour, and $[\text{CoCl}_4]^{2-}(\text{aq})$ which is blue. The equilibrium system is:



- a) If an aqueous solution of cobalt(II) chloride is placed in an ice water bath, the solution will turn blue. If the solution is placed in a hot water bath at 100°C , it will turn pink. Is the reaction as written above exothermic or endothermic? Explain your reasoning. (2 marks)

When heated, reaction shifts to the left (turns pink)

As exothermic reaction are heated, they shift left, making

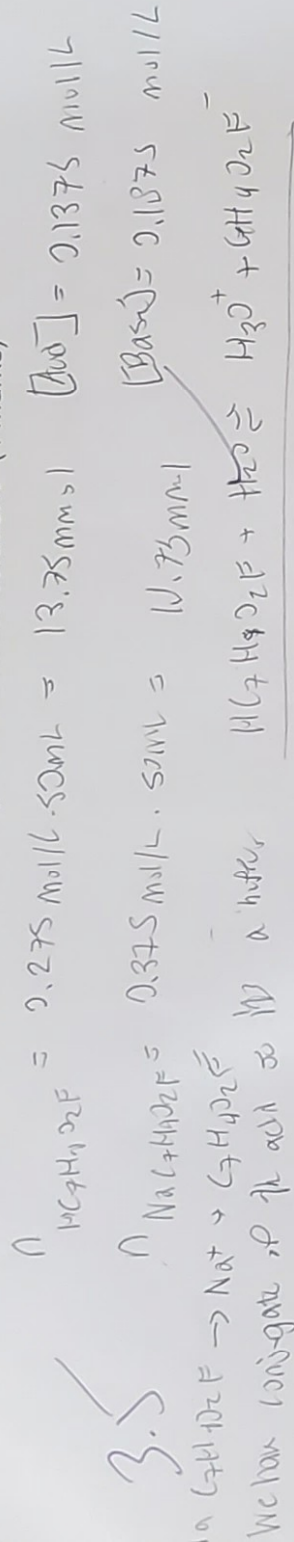
reaction exothermic. To compensate for the heat, it shifted left making it

- b) Explain how the addition of silver ions would affect this equilibrium? (1 mark)

It would react with the Cl^- making it shift left to compensate

for the loss of Cl^- .

2. A buffer solution is prepared by mixing 50.0 mL of 0.275 mol/L fluorobenzoic acid ($C_7H_5O_2F$) with 50.0 mL of 0.375 mol/L sodium fluorobenzoate. Show the chemical equation for the equilibrium system of this buffer. The pK_a of this weak acid is 2.90. What is the pH of the buffer solution? (4 marks)

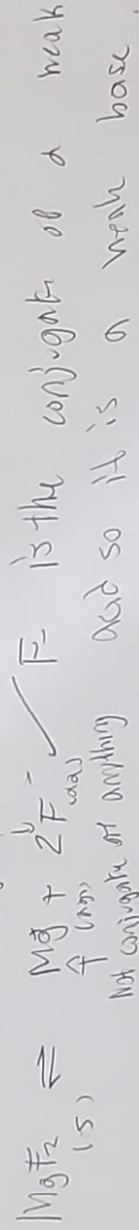


Using Hasselbach

$$pH = pK_a + \log_{10} \left(\frac{[C_7H_4O_2F^-]}{[C_7H_5O_2F]} \right)$$

$$pH = 2.9 + \log_{10} \left(\frac{0.1375}{0.1875} \right) = 3.035$$

3. Determine whether a solution of magnesium fluoride, MgF_2 , is acidic, basic or neutral. (3 marks)

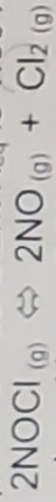


Since OH^- conjugate base is present, the solution will be slightly basic.

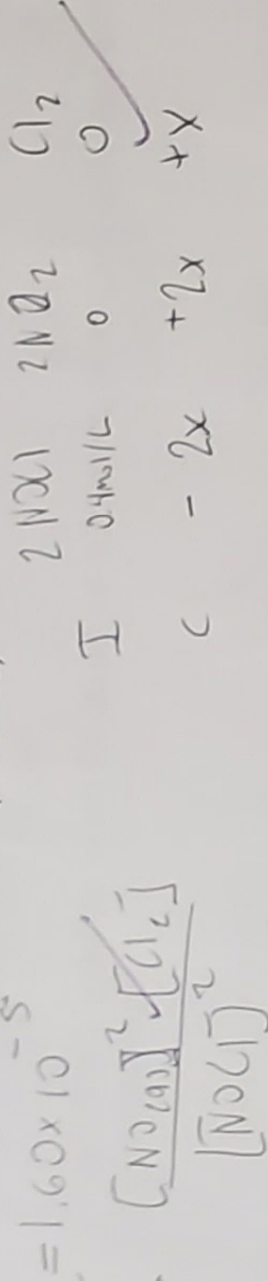
3



For the following reaction, the equilibrium constant K_{eq} is 1.60×10^{-5} at 35°C .



If 0.80 moles of $\text{NOCl}_{(g)}$ is introduced into a 2.00 L vessel, calculate the equilibrium concentration $\text{NO}_{(g)}$ at this temperature. (6 marks)



$$E \quad 0.4-2x \quad 2x \quad x$$

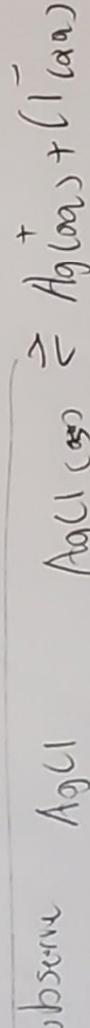
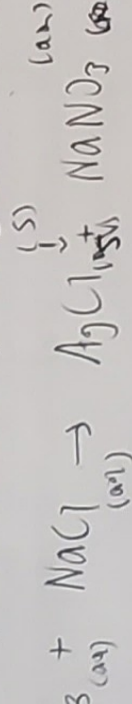
$$K \cdot 2 = [\text{NO}_{(g)}^2]$$

$$= \frac{4x^2}{(0.4-2x)^2} = 1.60 \times 10^{-5}$$

$$= 1.60 \times 10^{-5}$$

0.0172 mol/L is final concentration
2 S.D.

Will a precipitate form by mixing 50.0 mL of 0.12 M AgNO_3 with 50.0 mL of 0.048 M NaCl ? (5 marks)



$$K_{sp} = [\text{Ag}^+][\text{Cl}^-] = 1.8 \times 10^{-10}$$

$$= [0.06 \text{ mol/L}] [0.024 \text{ mol/L}]$$

$$= 0.00144$$

Compare with $K_{sp} = 1.8 \times 10^{-10}$

$Q > K_{sp}$ so precipitate will form

$$[\text{Ag}^+] = \frac{0.012}{100 \text{ mL}} = 0.06 \text{ mol/L}$$

$$= 0.06 \text{ mol/L}$$

$$[\text{Cl}^-] = \frac{0.024 \text{ mol/L}}{\text{precipitate forms}}$$